

Operations Research

BCA — Semester 4 — Answer Sheet

Subject Code:

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Group B

1. Solve the given Linear Programming Problem (LPP) by using simplex method and interpret the results. Minimize $Z = 2A + 10B$ Subject to constraints: $A + B = 50$, $A \geq 20$, $B \leq 40$ Where, $A, B \geq 0$

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

2. Find transport schedule to minimize the transportation cost for the following transportation problem. The transportation cost per unit and units demanded and available are given in the table.

From \ To	ABC	Units demanded
X	9	10
Y	10	5
Z	14	20
Units available	20	15

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. The table give below give the information about the activities, their predecessors and time duration required to complete the activities of the project.

Activity	Predecessor	Time in weeks
A		5
B	A	12
C	A	6
D	B, C	3
E	B, C	2
F	D, E	1
G	D, E	4
H	F, G	2

Draw the network diagram and identify the critical activities and critical path. Also find the minimum time duration required to complete the project.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

4. Callenoids Metropolitan is putting up bids for four used motorbikes company. The Metropolitan allows individuals to make bids on all four motorbikes company but will accept only one bid per individual have made the following bids (in thousands Rs).

Motorbike Company	Individuals	Bids
Honda	A	100
Hero	B	110
Bajaj	C	100
Yamaha	D	115

Make the use of Hungarian method to assign the individuals to different motorbike company in order to maximize the revenue.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. A milk salesman estimates the probability of the demand for a litre of milk is as follows:

Demand	Probability
1	0.1
2	0.3
3	0.2
4	0.1
5	0.3

He purchase a litre of milk @ of Rs. 60 and sells it @ of Rs. 70. Prepare payoff table and find the optimal act based on using EMV and EOL criterion determining the unsold milk has value.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. On the average 96 patients per 24 hours day require the emergency service in clinic. Also on the average, a patient requires 10 minutes of active attention. Assume that the facility can handle only one emergency at a time. If this situation satisfy the all the conditions for apply queuing theory, find the average (expected) queue length and the waiting time for the patient to be served.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. Determine the best strategy for each player A and B and value of the game. Player A's strategy: Player B's strategy: B B B B A A 40 40 40 40 A 30 30 20 50 A 10 30 90 20

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. The following tables gives the three kinds of foods and three kinds of vitamin contained on them. Formulate objective function and constraint of LPP for minimizing the cost. Vitamin Food Daily Requirements F F F V 20 10 10 300 V 10 10 10 200 V 10 20 10 240 Cost per unit of food Rs. 20 Rs. 24 Rs. 18

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. Describe modified distribution (MODI) method used for testing the optimality of initial solution of transport problem.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. What is called a queue? Describe the operating characteristics of the single channel queuing model.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. The following activities must be completed in order to complete the project. Draw network diagram and establish relationship between the activities of the project. Activity P Q R S T U V W X Predecessor -- P, Q Q P R T, U S, U V, W

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

12. Write short note on: a. Scopes of operations research b. Dominance Rule Method in game theory

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.